SDGs High School Workshop: Green Asian Games

50-Min Interactive Task • Sustainable Mega-Events Ideation | 50分鐘高中雙語永續亞運提案工作坊

GROUP 1: ACCOMMODATION & LIVING

Focus & Challenge (English) Topic 01

Focus: Support athletes without building a massive temporary Athlete Village.
Challenge: Decentralized hotel stays lack unified dining, physical therapy, and team lounges.

Creative Ideas to Brainstorm

- **Hotel Coalitions:** Shared physical therapy & dining hubs across hotel clusters.
- **Pop-up Mobile Services:** EV trucks for mobile athlete care and recovery.
- **Smart Eco-Rooms:** Gamify water & electricity saving for athlete stays.

Decentralized Housing | Modular Hubs | Shared Services

焦點與挑戰 (中文對照) 主題 01

重點任務：不興建大型一次性選手村，仍能提供完美的選手生活與醫療後勤。
面臨挑戰：分散入住一般飯店，缺乏集中式團膳、物理治療防護與交誼空間。

創意發想思考點

- **綠色飯店聯盟：**在相鄰飯店區域間建立共享服務後勤網絡與防護站。
- **快閃後勤據點：**運用電動拖車建立可巡迴的選手防護與健康飲食備餐車。
- **智慧房間節能：**透過遊戲化獎勵機制，鼓勵選手入住期間省水省電。

分散式住宿 | 模組化據點 | 共享後勤

📌 **How to Evaluate Feasibility (可行性如何評估？):**
- *Cost & Tech (成本與技術):* Can it be built within budget & existing tech?
- *Logistics (後勤):* Can hotels organize shared services smoothly?

🌱 **How to Evaluate Carbon Effectiveness (有效性如何評估？):**
- *Carbon Cut (減碳量):* Does avoiding a permanent village save massive CO2?
- *Scale (涵蓋範圍):* Can it serve thousands of athletes efficiently?

📊 Problem-Solving Matrix: Place Your Ideas into 4 Quadrants (問題解決矩陣)

Feasibility (X-axis) vs. Effectiveness (Y-axis)

⭐ **Q1: Golden Solution (高可行 + 高減碳)**
Easy to implement & big carbon reduction! (首選方案：可行性高且減碳效益大)

🚀 **Q2: Long-Term Goal (低可行 + 高減碳)**
High impact, but needs high budget/tech. (長遠目標：減碳量大但技術/成本門檻高)

⚡ **Q3: Low-Hanging Fruit (高可行 + 低減碳)**
Easy to do, but low total carbon cut. (簡單小事：容易執行但總體減碳量較有限)

⚠️ **Q4: Reconsider (低可行 + 低減碳)**
Hard to execute & low impact. Skip this. (建議放棄：執行困難且減碳效果不佳)

💡 Group Output: Selected Golden Solution & 1-Min Pitch

1. **Problem:** "When athletes stay in scattered hotels, [main challenge] is a major issue."
2. **Golden Solution:** "Using the matrix, our group chose [Q1 idea] as our best solution."
3. **Why It Works:** "Its Feasibility is high because [reason], and it cuts carbon effectively by [reduction mechanism]."

📝 Brainstorming Notes (小組紀錄區):

- Selected Q1 Idea: _____
- Key Carbon Benefit: _____
- Speaker Name: _____

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GROUP 2: CATERING & FOOD WASTE

Focus & Challenge (English) Topic 02

Focus: Provide low-carbon, high-quality meals for tens of thousands of participants.
Challenge: Large events produce massive leftover food waste and single-use plastic trash.

Creative Ideas to Brainstorm

- **Low-Carbon Menu:** Design Farm-to-Table plant-forward athletic nutrition.
- **AI Surplus Planning:** Predict exact meal portions to eliminate leftover food waste.
- **Zero-Waste Dining:** Use compostable or edible tableware in spectator zones.

Farm-to-Table | AI Portion Prediction | Edible Tableware

焦點與挑戰 (中文對照) 主題 02

重點任務：為數萬名選手與觀眾提供低碳、優質且安全的餐飲服務。
面臨挑戰：大型賽事極易產生驚人的剩食與一次性餐具廢棄物。

創意發想思考點

- **低碳在地菜單：**設計符合運動員營養、由在地農場直達餐桌的低碳植物肉料理。
- **AI 智慧備餐：**運用預測系統精準調配份量，將多餘剩食轉贈在地社區。
- **零廢棄餐食：**在觀賽區引進「可食用餐具」或 100% 生物可堆肥資材。

在地食材 | AI 減糧廢 | 可堆肥餐具

📌 **How to Evaluate Feasibility (可行性如何評估？):**
- *Cost & Supply (成本與供應鏈):* Can local farms supply ingredients safely?
- *Hygiene (衛生法規):* Is food redistribution legal and safe?

🌱 **How to Evaluate Carbon Effectiveness (有效性如何評估？):**
- *Food Miles (食物里程):* How much CO2 is saved by local sourcing?
- *Landfill Reduction (垃圾減量):* Does it prevent single-use plastic waste?

📊 Problem-Solving Matrix: Place Your Ideas into 4 Quadrants (問題解決矩陣)

Feasibility (X-axis) vs. Effectiveness (Y-axis)

⭐ **Q1: Golden Solution (高可行 + 高減碳)**
Easy to implement & big carbon reduction! (首選方案：可行性高且減碳效益大)

🚀 **Q2: Long-Term Goal (低可行 + 高減碳)**
High impact, but needs high budget/tech. (長遠目標：減碳量大但技術/成本門檻高)

⚡ **Q3: Low-Hanging Fruit (高可行 + 低減碳)**
Easy to do, but low total carbon cut. (簡單小事：容易執行但總體減碳量較有限)

⚠️ **Q4: Reconsider (低可行 + 低減碳)**
Hard to execute & low impact. Skip this. (建議放棄：執行困難且減碳效果不佳)

💡 Group Output: Selected Golden Solution & 1-Min Pitch

1. **Problem:** "In large dining zones, food waste and packaging [main challenge]."
2. **Golden Solution:** "Using the matrix, our group chose [Q1 idea] as our best solution."
3. **Why It Works:** "Its Feasibility is high because [reason], and it cuts carbon effectively by [reduction mechanism]."

📝 **Brainstorming Notes (小組紀錄區):**

- Selected Q1 Idea: _____
- Key Carbon Benefit: _____
- Speaker Name: _____

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GROUP 3: MOBILITY & LOGISTICS

Focus & Challenge (English) Topic 03

Focus: Achieve zero-emission, punctual transportation across dispersed sites.
Challenge: Traffic congestion and vehicle idling significantly increase total emissions.

Creative Ideas to Brainstorm

- **Shared e-Mobility:** Integrate EV buses, hydrogen fleet, and public transit passes.
- **AI Fleet Dispatch:** Use dynamic routing to avoid congestion and vehicle idling.
- **Micro-Mobility Lanes:** Design green dedicated lanes for e-scooters and bicycles.

Dynamic Routing | Zero-Emission Fleet | Transit Pass Integration

焦點與挑戰 (中文對照) 主題 03

重點任務：場館與住宿分散時，實現零碳排、不延誤的精準接駁調度。
面臨挑戰：道路塞車與接駁車輛空轉（Idling）會產生大量的廢氣與排碳量。

創意發想思考點

- **共享電動車隊：**整合純電巴士、氫能車與觀賽者大眾運輸綠色 Pass。
- **AI 智慧車隊調度：**動態優化避塞路線，確保選手精準抵達且車輛不空跑。
- **微型移動專用道：**劃設專用綠色動線，提供電動滑板車供短程移動。

動態路線規劃 | 零碳車隊 | 綠色交通整合

📌 **How to Evaluate Feasibility (可行性如何評估？):**
- *Infrastructure (基礎設施):* Are charging/hydrogen stations available?
- *Punctuality (準點率):* Will athletes arrive on time without delay?

🌱 **How to Evaluate Carbon Effectiveness (有效性如何評估？):**
- *Tailpipe Emissions (廢氣減量):* Does replacing gas buses eliminate CO2?
- *Traffic Flow (車流優化):* Does AI dispatching stop engine idling?

📊 Problem-Solving Matrix: Place Your Ideas into 4 Quadrants (問題解決矩陣)

Feasibility (X-axis) vs. Effectiveness (Y-axis)

⭐ Q1: Golden Solution (高可行 + 高減碳)
Easy to implement & big carbon reduction! (首選方案：可行性高且減碳效益大)

🚀 Q2: Long-Term Goal (低可行 + 高減碳)
High impact, but needs high budget/tech. (長遠目標：減碳量大但技術/成本門檻高)

⚡ Q3: Low-Hanging Fruit (高可行 + 低減碳)
Easy to do, but low total carbon cut. (簡單小事：容易執行但總體減碳量較有限)

⚠️ Q4: Reconsider (低可行 + 低減碳)
Hard to execute & low impact. Skip this. (建議放棄：執行困難且減碳效果不佳)

💡 Group Output: Selected Golden Solution & 1-Min Pitch

1. **Problem:** "When venues are scattered, vehicle congestion [main challenge]."
2. **Golden Solution:** "Using the matrix, our group chose [Q1 idea] as our best solution."
3. **Why It Works:** "Its Feasibility is high because [reason], and it cuts carbon effectively by [reduction mechanism]."

📝 Brainstorming Notes (小組紀錄區):

- Selected Q1 Idea: _____
- Key Carbon Benefit: _____
- Speaker Name: _____

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GROUP 4: ENERGY & VENUE OPERATIONS

Focus & Challenge (English) Topic 04

Focus: Operate competition venues using 100% net-zero energy.
Challenge: Old stadiums consume high energy, and temporary setups create construction waste.

Creative Ideas to Brainstorm

- **Modular Venues:** Design 100% reusable, rented temporary stadium structures.
- **Renewable Microgrids:** Apply solar panels, green hydrogen flames, and storage.
- **Smart Cooling:** Use natural ventilation, green walls, and misting instead of AC.

Modular Construction | Green Hydrogen Flame | Passive Cooling

焦點與挑戰 (中文對照) 主題 04

重點任務：在不大量新建場館的前提下，使比賽營運過程達到淨零碳排。
面臨挑戰：既有舊場館耗能高，且臨時搭建的設施容易在賽後變成營建廢棄物。

創意發想思考點

- **模組化可拆場館：**採用可 100% 拆解租賃、賽後轉作其他用途的設施。
- **再生能源微電網：**導入太陽能光電板、綠氫聖火與儲能設備。
- **智慧降溫技術：**利用建築通風、綠牆與霧化降溫，減少高耗能冷氣使用。

可拆解建築 | 綠氫聖火 | 被動式降溫

📌 **How to Evaluate Feasibility (可行性如何評估？):**
- *Power Grid (電網穩定度):* Can solar/storage supply continuous power?
- *Construction (施工難度):* Is modular structure easy to set up & remove?

🌱 **How to Evaluate Carbon Effectiveness (有效性如何評估？):**
- *Embodied Carbon (隱含碳):* Avoids concrete/steel building carbon?
- *Clean Power (綠電比例):* Does it achieve 100% clean power?

📊 Problem-Solving Matrix: Place Your Ideas into 4 Quadrants (問題解決矩陣)

Feasibility (X-axis) vs. Effectiveness (Y-axis)

⭐ **Q1: Golden Solution (高可行 + 高減碳)**
Easy to implement & big carbon reduction! (首選方案：可行性高且減碳效益大)

🚀 **Q2: Long-Term Goal (低可行 + 高減碳)**
High impact, but needs high budget/tech. (長遠目標：減碳量大但技術/成本門檻高)

⚡ **Q3: Low-Hanging Fruit (高可行 + 低減碳)**
Easy to do, but low total carbon cut. (簡單小事：容易執行但總體減碳量較有限)

⚠️ **Q4: Reconsider (低可行 + 低減碳)**
Hard to execute & low impact. Skip this. (建議放棄：執行困難且減碳效果不佳)

💡 Group Output: Selected Golden Solution & 1-Min Pitch

1. **Problem:** "Traditional venues consume heavy energy during [main challenge]."
2. **Golden Solution:** "Using the matrix, our group chose [Q1 idea] as our best solution."
3. **Why It Works:** "Its Feasibility is high because [reason], and it cuts carbon effectively by [reduction mechanism]."

📝 Brainstorming Notes (小組紀錄區):

- Selected Q1 Idea: _____
- Key Carbon Benefit: _____
- Speaker Name: _____

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GROUP 5: DIGITALIZATION & FAN ENGAGEMENT

Focus & Challenge (English) Topic 05

Focus: Use digital tech to cut paper waste and empower fans to take climate action.
Challenge: Printed tickets, guidebooks, and spectator travel generate high emissions.

Creative Ideas to Brainstorm

- **100% Paperless Event:** Mobile ticketing, AR navigation, and digital manuals.
- **Fan Carbon Rewards:** Gamify public transit use to earn digital souvenirs.
- **Virtual Spectator Zone:** Reduce flight emissions through immersive VR viewing.

Paperless System | Carbon Gamification | Virtual Fan Zone

焦點與挑戰 (中文對照) 主題 05

重點任務：透過數位科技減少紙張消耗，並帶動百萬觀賽群眾一起參與減碳。
面臨挑戰：實體門票、印刷手冊與觀眾遠途交通帶來極高的碳足跡。

創意發想思考點

- **100% 無紙化大會：**全面使用電子票券、AR 現場導覽與數位賽事手冊。
- **觀眾減碳獎勵：**搭捷運觀賽即可累積「減碳點數」，兌換限定數位徽章。
- **線上虛擬觀賽：**提供沉浸式 VR 觀賽體驗，減少跨國飛行產生的碳排。

無紙化系統 | 減碳遊戲化 | 虛擬觀賽區

📌 How to Evaluate Feasibility (可行性如何評估？):
- *App Adoption (使用者普及率):* Is the app user-friendly for all ages?
- *Digital Barrier (數位門檻):* Does VR require expensive devices?

🌱 How to Evaluate Carbon Effectiveness (有效性如何評估？):
- *Paper Saved (減紙效益):* Prevents millions of printed guidebooks?
- *Travel Cut (交通減碳):* Prompts fans to take eco-friendly transit?

📊 Problem-Solving Matrix: Place Your Ideas into 4 Quadrants (問題解決矩陣)

Feasibility (X-axis) vs. Effectiveness (Y-axis)

⭐ Q1: Golden Solution (高可行 + 高減碳)
Easy to implement & big carbon reduction! (首選方案：可行性高且減碳效益大)

🚀 Q2: Long-Term Goal (低可行 + 高減碳)
High impact, but needs high budget/tech. (長遠目標：減碳量大但技術/成本門檻高)

⚡ Q3: Low-Hanging Fruit (高可行 + 低減碳)
Easy to do, but low total carbon cut. (簡單小事：容易執行但總體減碳量較有限)

⚠️ Q4: Reconsider (低可行 + 低減碳)
Hard to execute & low impact. Skip this. (建議放棄：執行困難且減碳效果不佳)

💡 Group Output: Selected Golden Solution & 1-Min Pitch

1. **Problem:** "Paper tickets and fan travel result in [main challenge]."
2. **Golden Solution:** "Using the matrix, our group chose [Q1 idea] as our best solution."
3. **Why It Works:** "Its Feasibility is high because [reason], and it cuts carbon effectively by [reduction mechanism]."

📝 Brainstorming Notes (小組紀錄區):

- Selected Q1 Idea: _____
- Key Carbon Benefit: _____
- Speaker Name: _____

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GROUP 6: LEGACY & CIRCULAR ECONOMY

Focus & Challenge (English) Topic 06

Focus: Ensure 100% material circularity and leave a lasting ecological legacy.
Challenge: Post-game equipment, uniforms, and podiums often turn into unwanted waste.

Creative Ideas to Brainstorm

- **Upcycled Merchandising:** Make medals and uniforms from e-waste & ocean plastic.
- **Equipment Donation:** Donate sports gear to rural schools and local sports clubs.
- **Blue Carbon Offsetting:** Offset flight emissions via mangrove restoration projects.

Upcycled Medals | Equipment Recycling | Blue Carbon Credits

焦點與挑戰 (中文對照) 主題 06

重點任務：賽事結束後，所有資材與設備 100% 循環利用，留下永續資產。
面臨挑戰：賽事結束後，大量運動器材、服裝與頒獎台易變成廢棄物。

創意發想思考點

- **海廢與回收再製：**獎牌與志工服全面採用舊手機廢棄物與海洋塑膠製作。
- **器材捐贈計畫：**比賽用具（球類、計分板）賽後全數捐贈給偏鄉學校。
- **藍碳復育抵銷：**無法避免的航空碳排，透過投資海洋紅樹林進行碳抵銷。

廢棄物再製獎牌 | 資材捐贈 | 藍碳抵銷

📌 **How to Evaluate Feasibility (可行性如何評估？):**
- *Recycling Tech (回收技術):* Are local upcycling factories ready?
- *Community Link (社區對接):* Is there a clear path for donations?

🌱 **How to Evaluate Carbon Effectiveness (有效性如何評估？):**
- *Circular Value (循環價值):* Prevents gear from going to incinerators?
- *Carbon Sequestration (固碳量):* Does mangrove planting absorb CO2?

📊 Problem-Solving Matrix: Place Your Ideas into 4 Quadrants (問題解決矩陣)

Feasibility (X-axis) vs. Effectiveness (Y-axis)

⭐ **Q1: Golden Solution (高可行 + 高減碳)**
Easy to implement & big carbon reduction! (首選方案：可行性高且減碳效益大)

🚀 **Q2: Long-Term Goal (低可行 + 高減碳)**
High impact, but needs high budget/tech. (長遠目標：減碳量大但技術/成本門檻高)

⚡ **Q3: Low-Hanging Fruit (高可行 + 低減碳)**
Easy to do, but low total carbon cut. (簡單小事：容易執行但總體減碳量較有限)

⚠️ **Q4: Reconsider (低可行 + 低減碳)**
Hard to execute & low impact. Skip this. (建議放棄：執行困難且減碳效果不佳)

💡 Group Output: Selected Golden Solution & 1-Min Pitch

1. **Problem:** "Post-game equipment and uniforms often become [main challenge] ."
2. **Golden Solution:** "Using the matrix, our group chose [Q1 idea] as our best solution."
3. **Why It Works:** "Its Feasibility is high because [reason] , and it cuts carbon effectively by [reduction mechanism] ."

📝 Brainstorming Notes (小組紀錄區):

- Selected Q1 Idea: _____
- Key Carbon Benefit: _____
- Speaker Name: _____